

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

- LO 3.10.A** — Solve a trigonometric equation by isolating the trig function and using inverse trig to find all solutions. **Isolate trig function → apply inverse → find both solutions in one period → add $+2\pi k$.**
- LO 3.10.A** — Write the general solution of a trig equation using periodicity, and apply a domain restriction when context requires it. **General solution uses $+2\pi k$; context limits which k -values are valid.**

Vocabulary & Key Concepts

trigonometric equation: An equation in which the unknown appears inside a **trigonometric** function.

principal solution: The solution obtained by applying an **inverse** trig function directly; limited to the **range** of that inverse function.

general solution: All solutions, expressed by adding integer multiples of the **period** to the principal value(s).
Example: $\theta = \frac{\pi}{6} + 2\pi k, k \in \mathbb{Z}$.

domain restriction: A limit on the variable imposed by the **context** of the problem (e.g. $0 \leq t \leq 24$ for a 24-hour model).

LO 3.10.A — EK 3.10.A.2: Infinitely Many Solutions

Key Idea: Because trig functions are **periodic**, a trig equation typically has **infinitely** many solutions when the domain is all real numbers.

Example: Solve $\sin \theta = \frac{1}{2}$.

Step 1 — Principal value: $\theta = \arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$

Step 2 — Second solution in $[0, 2\pi)$: Sine is positive in Quadrants **I** and **II**. Reflect: $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

Step 3 — General solution: Add multiples of the period 2π to each solution.

Family 1: $\theta = \frac{\pi}{6} + 2\pi k$ Family 2: $\theta = \frac{5\pi}{6} + 2\pi k$

where k is any **integer**.

Check: List 3 specific solutions from each family:

Family 1 ($k = 0, 1, 2$): $\frac{\pi}{6}, \frac{\pi}{6} + 2\pi = \frac{13\pi}{6}, \frac{\pi}{6} + 4\pi = \frac{25\pi}{6}$

Family 2 ($k = 0, 1, -1$): $\frac{5\pi}{6}, \frac{5\pi}{6} + 2\pi = \frac{17\pi}{6}, \frac{5\pi}{6} - 2\pi = -\frac{7\pi}{6}$

LO 3.10.A — EK 3.10.A.1: The Four-Step Procedure

Algorithm for solving a trig equation:

- Isolate** the trig function on one side of the equation.
- Apply** the inverse trig function to find the principal value.

3. **Find** the second solution in $[0, 2\pi)$ using reference angles and the appropriate quadrant(s).
4. **Write** the general solution by adding $+2\pi k$ (or $+\pi k$ for tangent) to each solution.

Example: Solve $2 \cos \theta - 1 = 0$.

Step 1 — Isolate: $\cos \theta = \frac{1}{2}$

Step 2 — Principal value: $\theta = \arccos(\frac{1}{2}) = \frac{\pi}{3}$

Step 3 — Cosine is positive in **QI** and **QIV**. Second solution: $\theta = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

Step 4 — General solution:

$$\theta = \frac{\pi}{3} + 2\pi k \text{ or } \theta = \frac{5\pi}{3} + 2\pi k, k \in \mathbb{Z}$$

Solutions in $[0, 2\pi)$: $\theta = \frac{\pi}{3}$ **and** $\theta = \frac{5\pi}{3}$

LO 3.10.A — EK 3.10.A.3: Contextual Domain Restrictions

Context: The height of water at a harbor (in feet) is modeled by $h(t) = 3 \sin(\frac{\pi}{6}t) + 5$, where t is hours after midnight, $0 \leq t \leq 24$.

Question: When is $h(t) \geq 7$?

Step 1 — Rewrite: $3 \sin(\frac{\pi}{6}t) + 5 \geq 7 \Rightarrow \sin(\frac{\pi}{6}t) \geq \frac{2}{3}$

Step 2 — Let $u = \frac{\pi}{6}t$. Solve $\sin u = \frac{2}{3}$ for the boundary:

$$u_1 = \arcsin(\frac{2}{3}) \approx 0.7297 \text{ rad} \quad u_2 = \pi - \arcsin(\frac{2}{3}) \approx 2.4119 \text{ rad}$$

Step 3 — Solve for t : $t = \frac{6}{\pi} \cdot u$

$$t_1 \approx 1.39 \text{ hr} \quad t_2 \approx 4.61 \text{ hr}$$

Step 4 — The inequality $h(t) \geq 7$ holds on the interval $1.39 \leq t \leq 4.61$.

Interpret: The ship can enter between approximately **1.39** and **4.61** hours after midnight.

Why do we restrict to $[0, 24]$? **The model represents one 24-hour period; times outside $[0, 24]$ have no physical meaning in this context.**

Guided Practice

Guided Practice 1. Solve $\sqrt{2} \sin \theta - 1 = 0$ for all $\theta \in [0, 2\pi)$. Show the four steps.

Step 1: $\sin \theta = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. **Step 2:** $\theta = \arcsin(\frac{\sqrt{2}}{2}) = \frac{\pi}{4}$. **Step 3:** Sine positive in **Q1** and **Q2**; second solution $\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$. **Step 4:** Solutions in $[0, 2\pi)$ are $\theta = \frac{\pi}{4}$ and $\theta = \frac{3\pi}{4}$.

Guided Practice 2. Write the general solution to $\cos \theta = -\frac{\sqrt{3}}{2}$.

$$\theta = \frac{5\pi}{6} + 2\pi k \text{ or } \theta = \frac{7\pi}{6} + 2\pi k, k \in \mathbb{Z}$$

Teacher Note

$\cos \theta = -\frac{\sqrt{3}}{2}$: reference angle is $\frac{\pi}{6}$; cosine is negative in Q2 and Q3, giving $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$ as the two solutions in $[0, 2\pi)$.

Guided Practice 3. A temperature model gives $T(t) = 10 \sin(\frac{\pi}{12}t) + 65$ for $t \in [0, 24]$ (hours). For what hours is $T(t) \geq 70$?

Set up the inequality and isolate the trig function:

$$10 \sin\left(\frac{\pi}{12}t\right) + 65 \geq 70 \Rightarrow \sin\left(\frac{\pi}{12}t\right) \geq \frac{1}{2}$$

Solve for the boundary values of t :

$\frac{\pi}{12}t = \frac{\pi}{6}$ **or** $\frac{\pi}{12}t = \frac{5\pi}{6}$, **so** $t = 2$ **or** $t = 10$. **Sine** $\geq \frac{1}{2}$ **between the two boundary values.**

Answer: $T(t) \geq 70$ on $2 \leq t \leq 10$.